

BORDER OPTICS

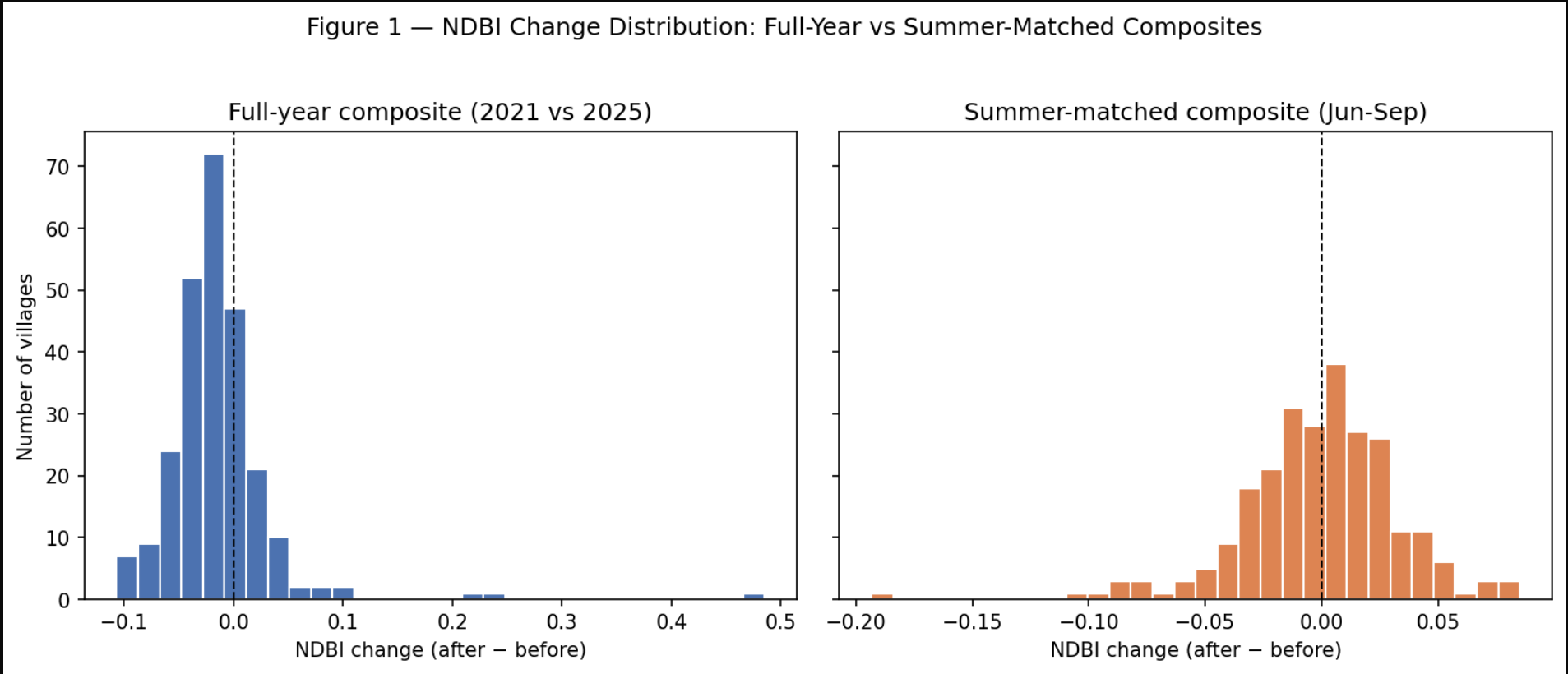
Maps & Plots

Satellite-Based Verification Framework for India's Vibrant Villages Programme (VVP-I)

Contents

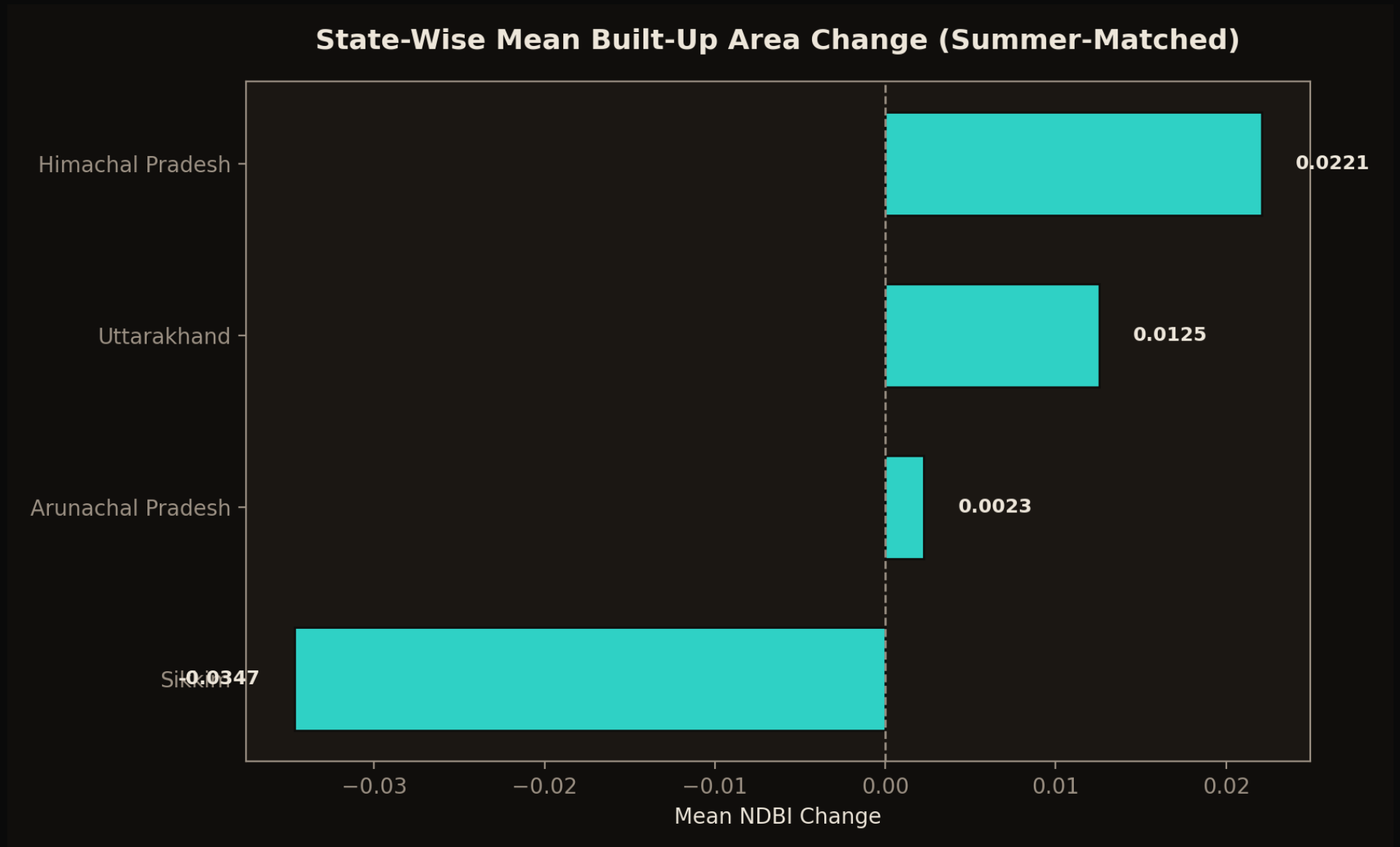
1. Figure 1 — NDBI Change Distribution
2. Figure 2 — State-Wise Mean Built-Up Area Change
3. Figure 3 — Distribution of Night-Lights Change
4. Figure 4 — State-Wise Mean Night-Lights Change
5. Figure 5 — State-Level Built-up Change vs Sanctioned Budget
6. Figure 6 — H3: Border Distance vs NDBI and Night-Light Change
7. Figure 7 — Control-Group Difference-in-Differences Effect
8. Figure 8 — Three-Point Multi-Year Trend
9. Figure 9 — Buffer-Radius Sensitivity
10. Figure 10 — Robustness Summary
11. Map 1 — Village Sample Overview
12. Map 2 — NDBI Change (Full-Year Composite)
13. Map 3 — NDBI Change (Summer-Matched Composite)
14. Map 4 — Night-Lights Change
15. Map 5 — Village Proximity to Border/LAC

Figure 1 — NDBI Change Distribution



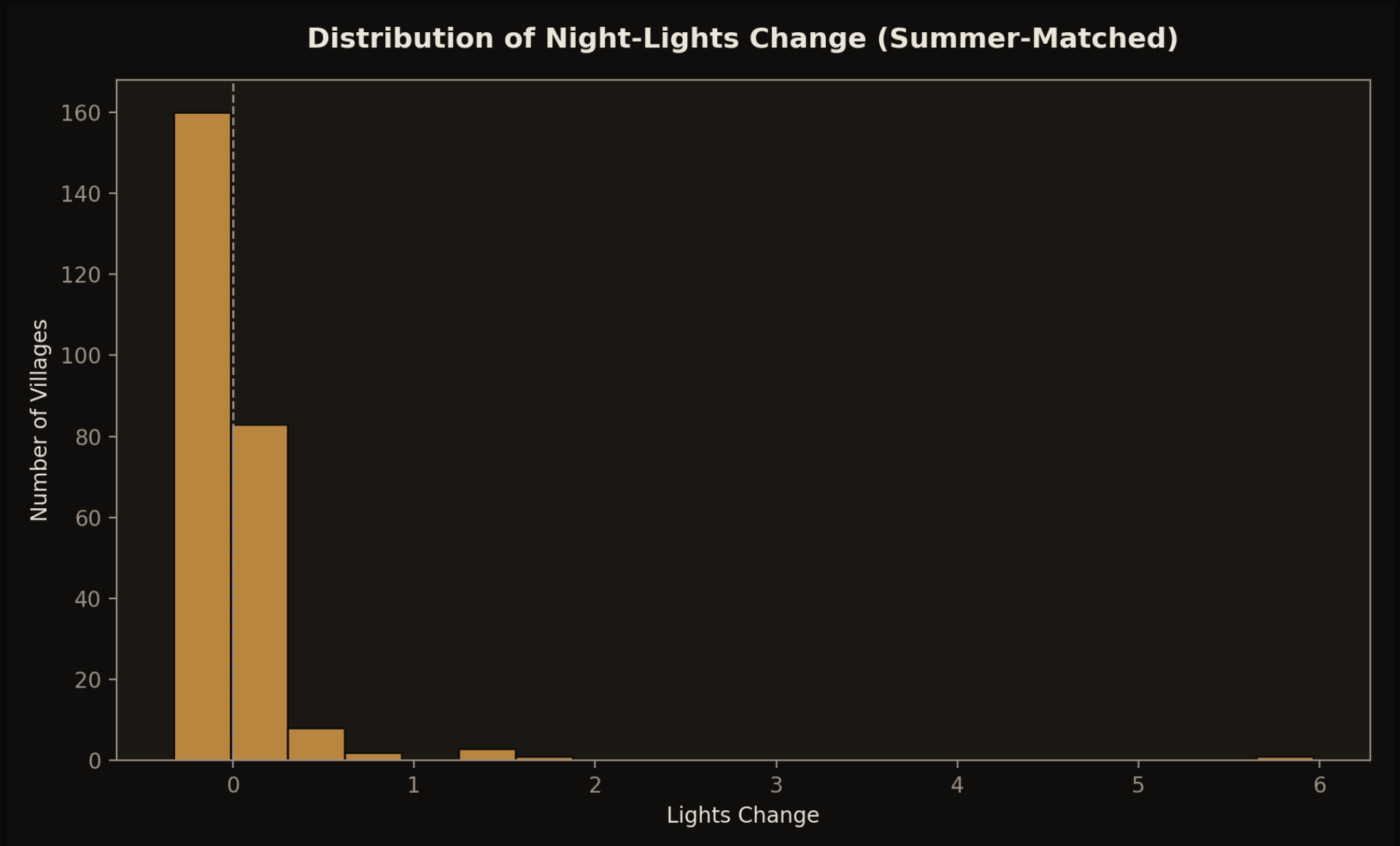
Full-year vs summer-matched composites, all geocoded villages.

Figure 2 — State-Wise Mean Built-Up Area Change



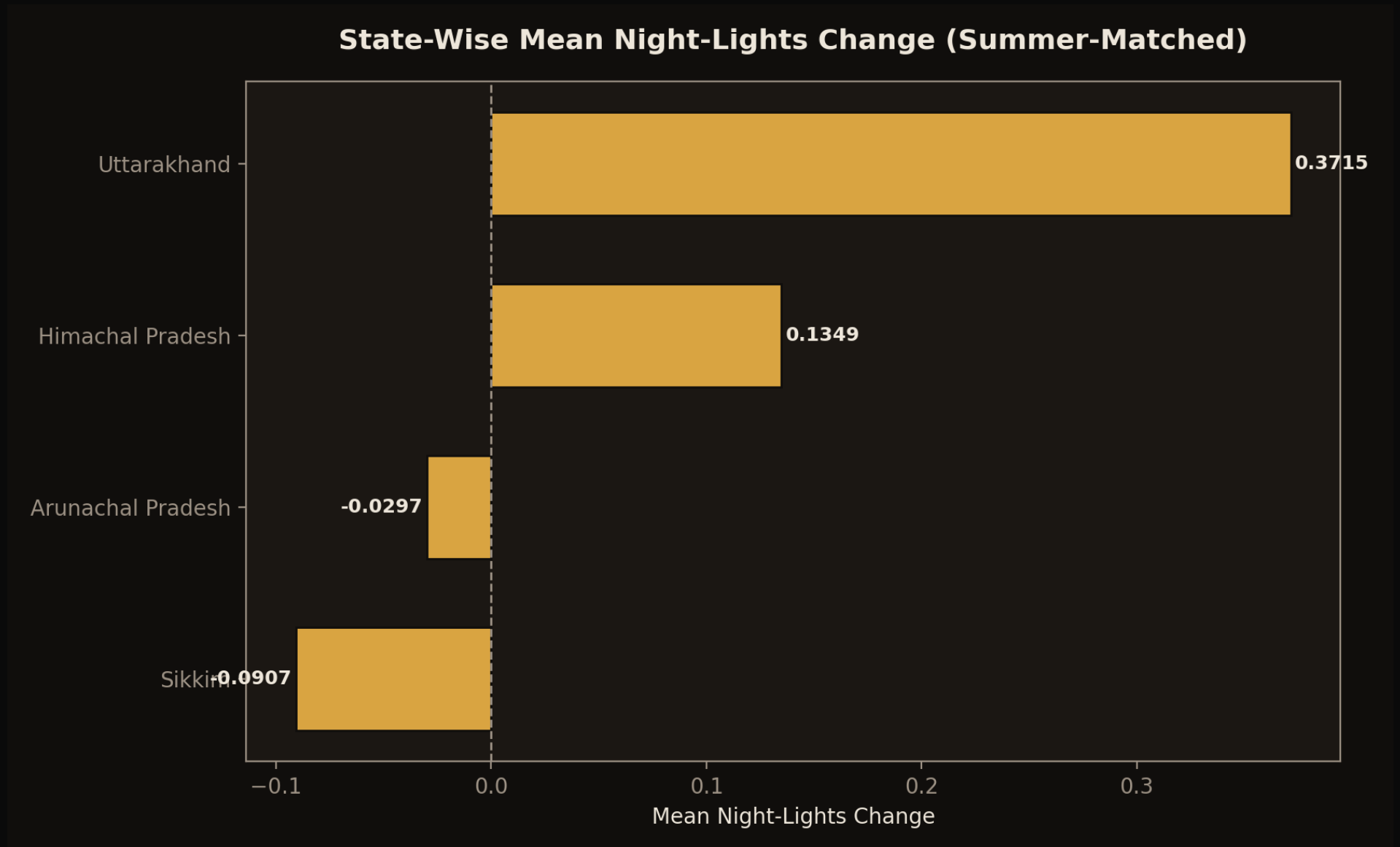
Summer-matched composite, by state.

Figure 3 — Distribution of Night-Lights Change



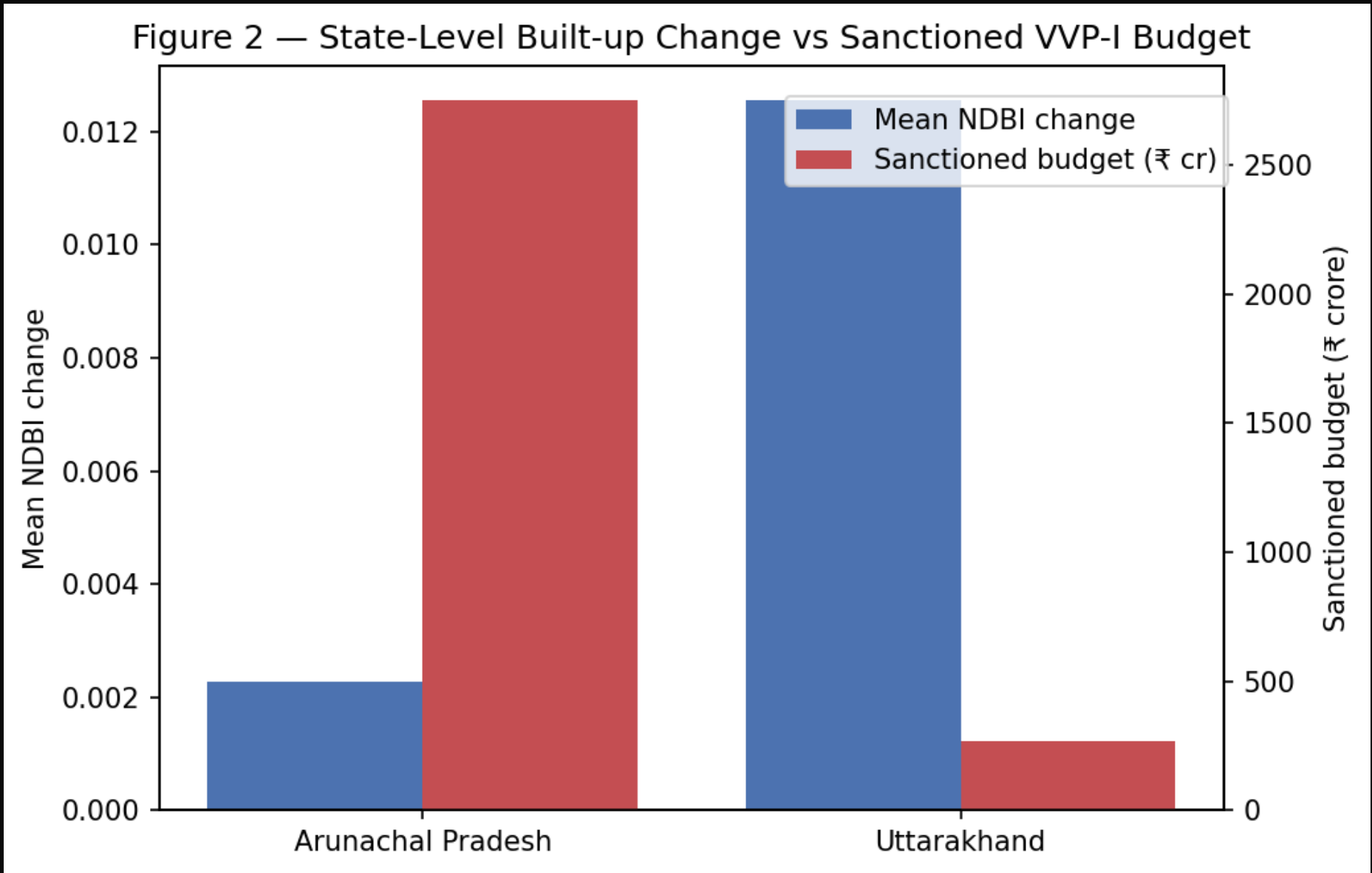
Summer-matched composite, all geocoded villages.

Figure 4 — State-Wise Mean Night-Lights Change



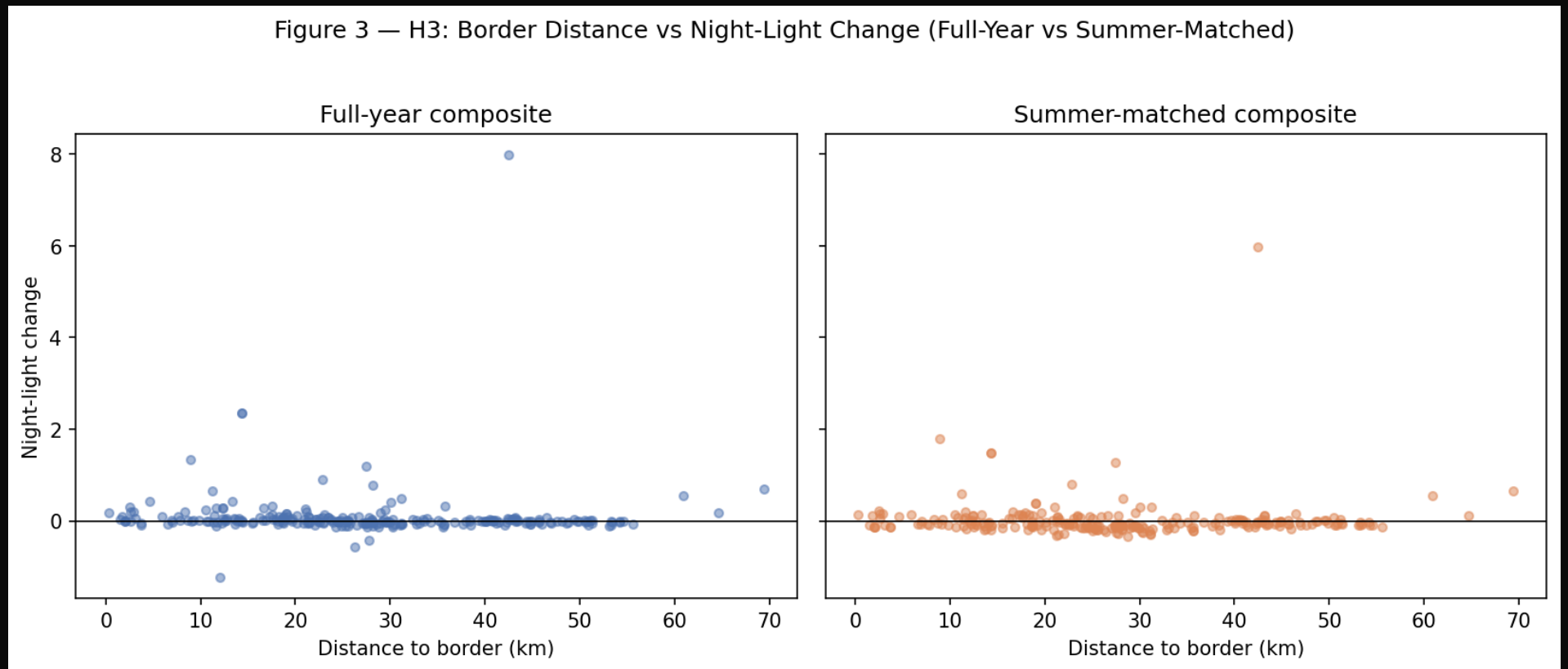
Summer-matched composite, by state.

Figure 5 — State-Level Built-up Change vs Sanctioned Budget



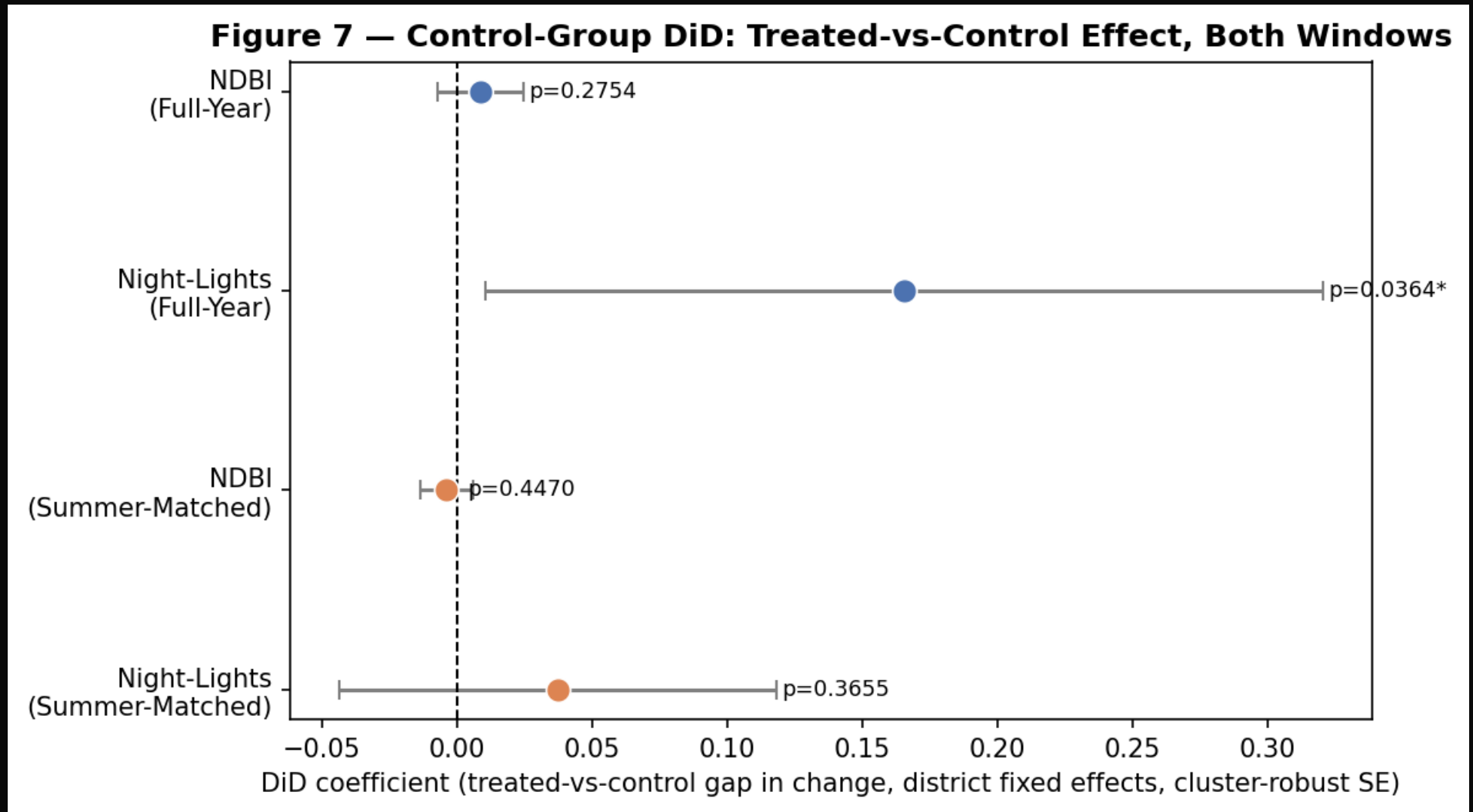
Summer-matched NDBI change, Arunachal Pradesh vs Uttarakhand (RQ2).

Figure 6 — H3: Border Distance vs NDBI and Night-Light Change



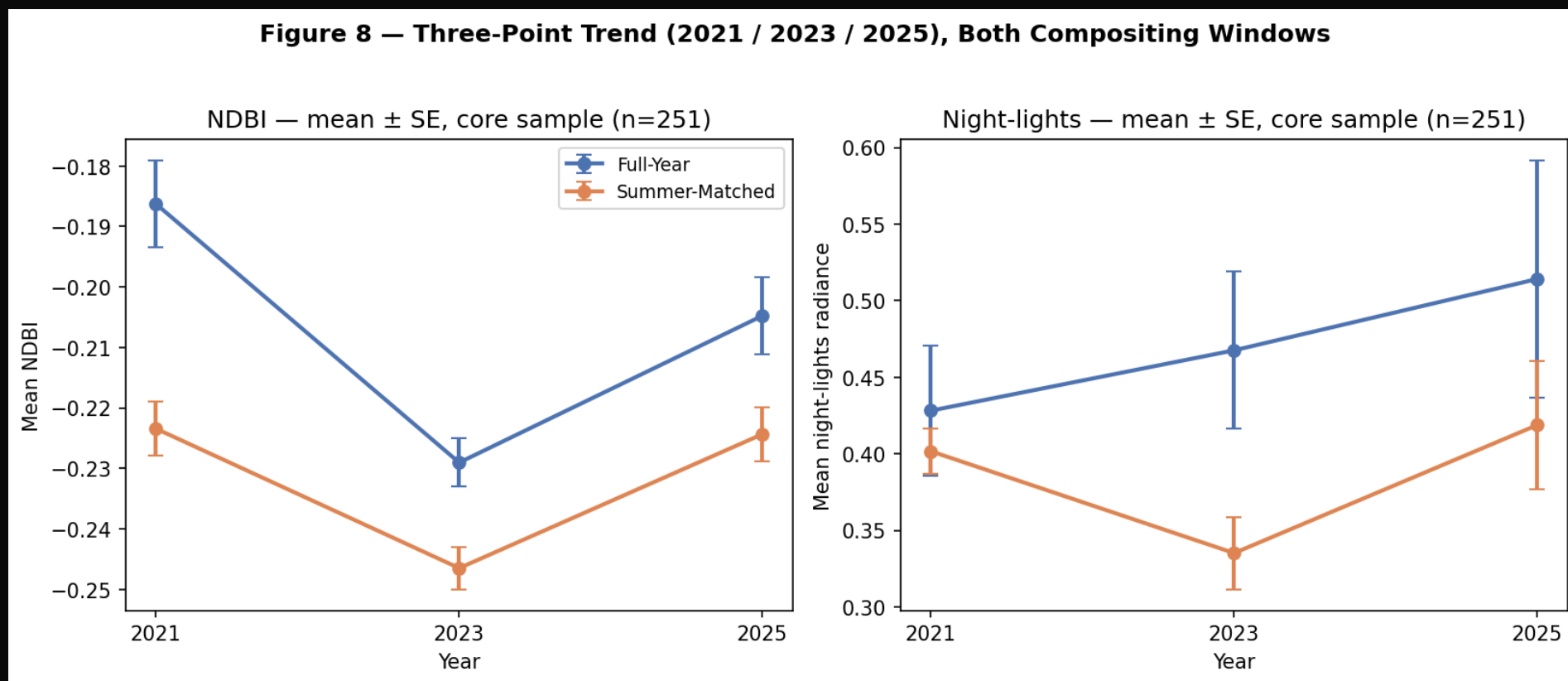
Full-year vs summer-matched composites.

Figure 7 — Control-Group Difference-in-Differences Effect



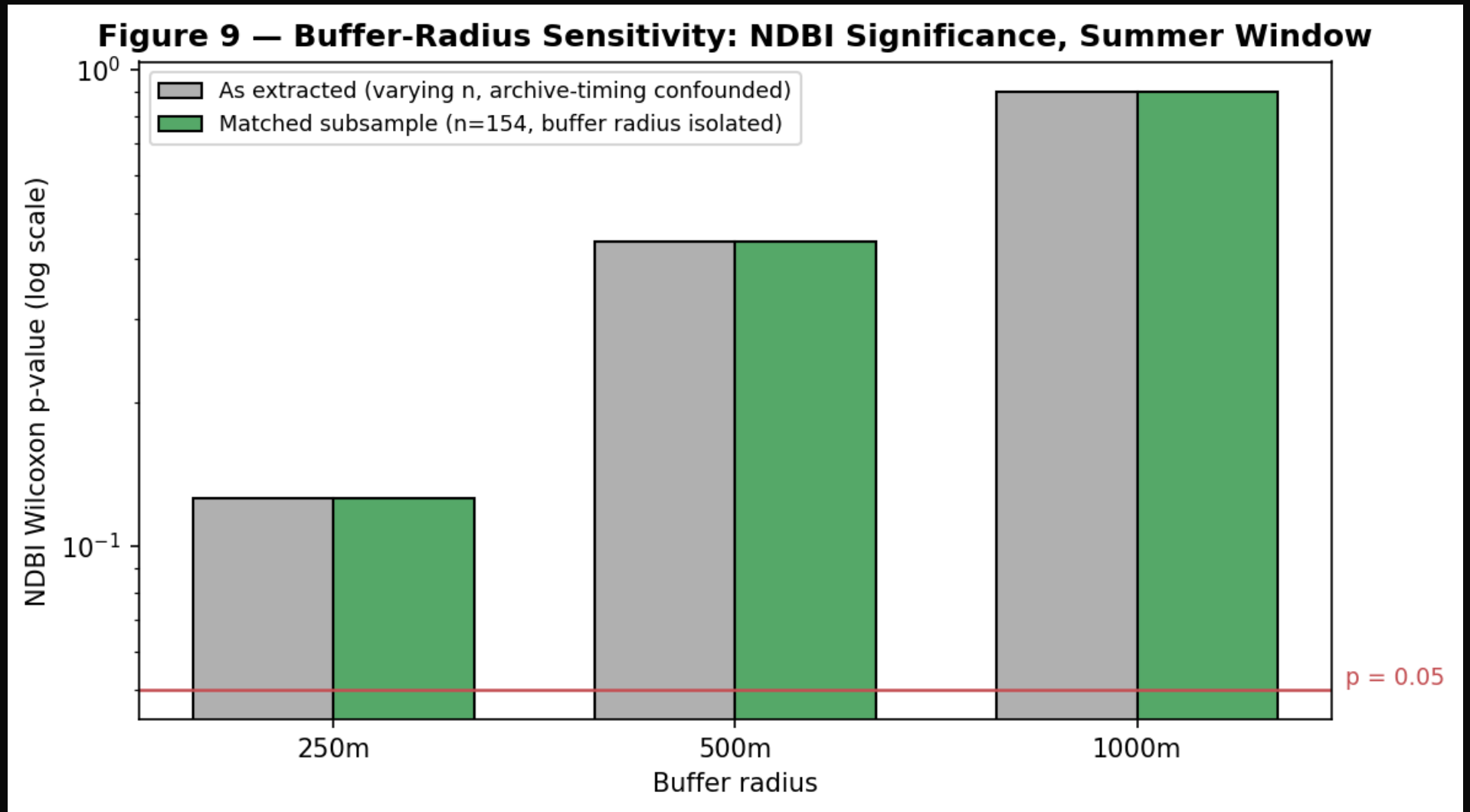
Treated-vs-control gap in change, district fixed effects, both compositing windows (H4).

Figure 8 — Three-Point Multi-Year Trend



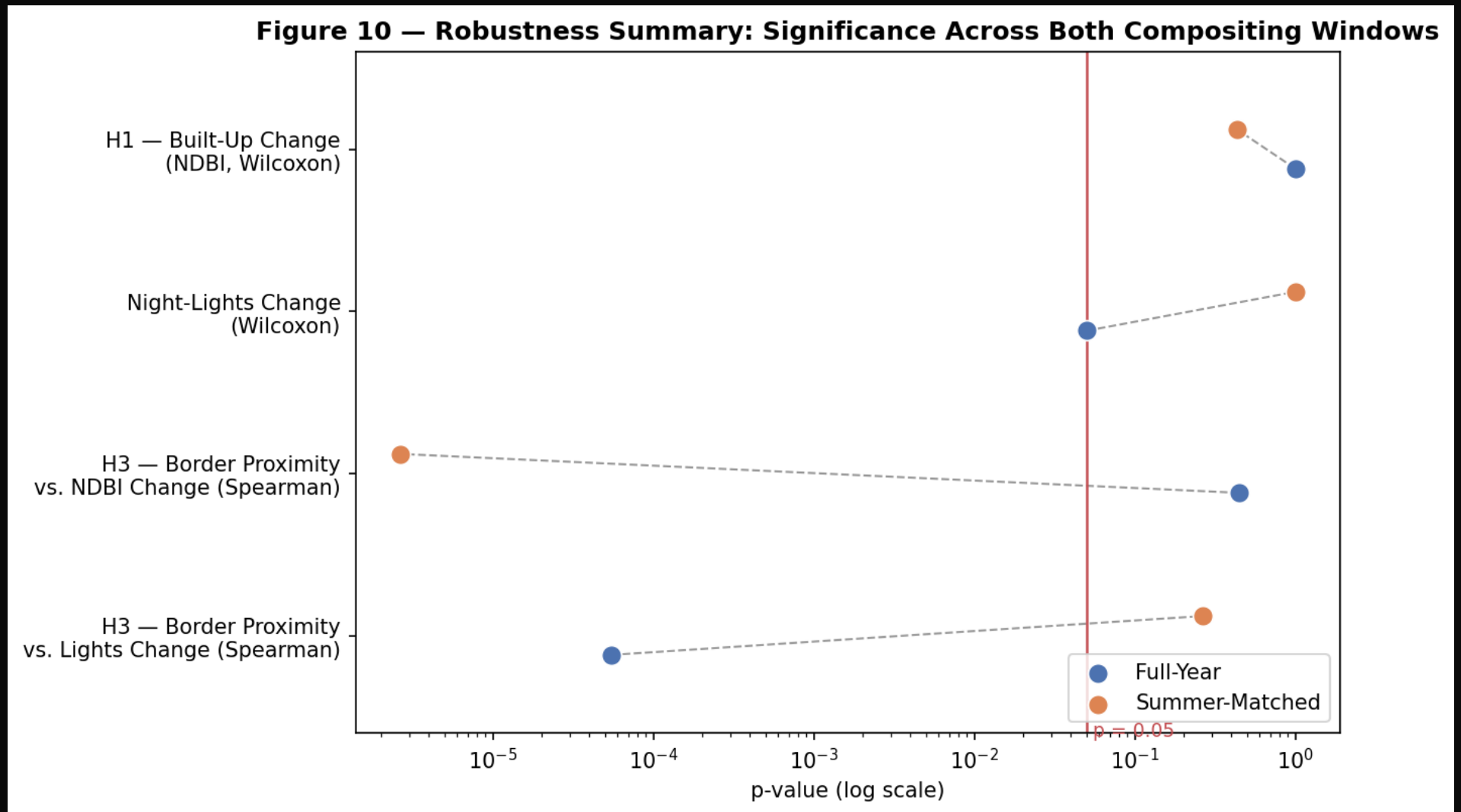
Mean NDBI and night-lights radiance at 2021, 2023, and 2025, both compositing windows.

Figure 9 — Buffer-Radius Sensitivity



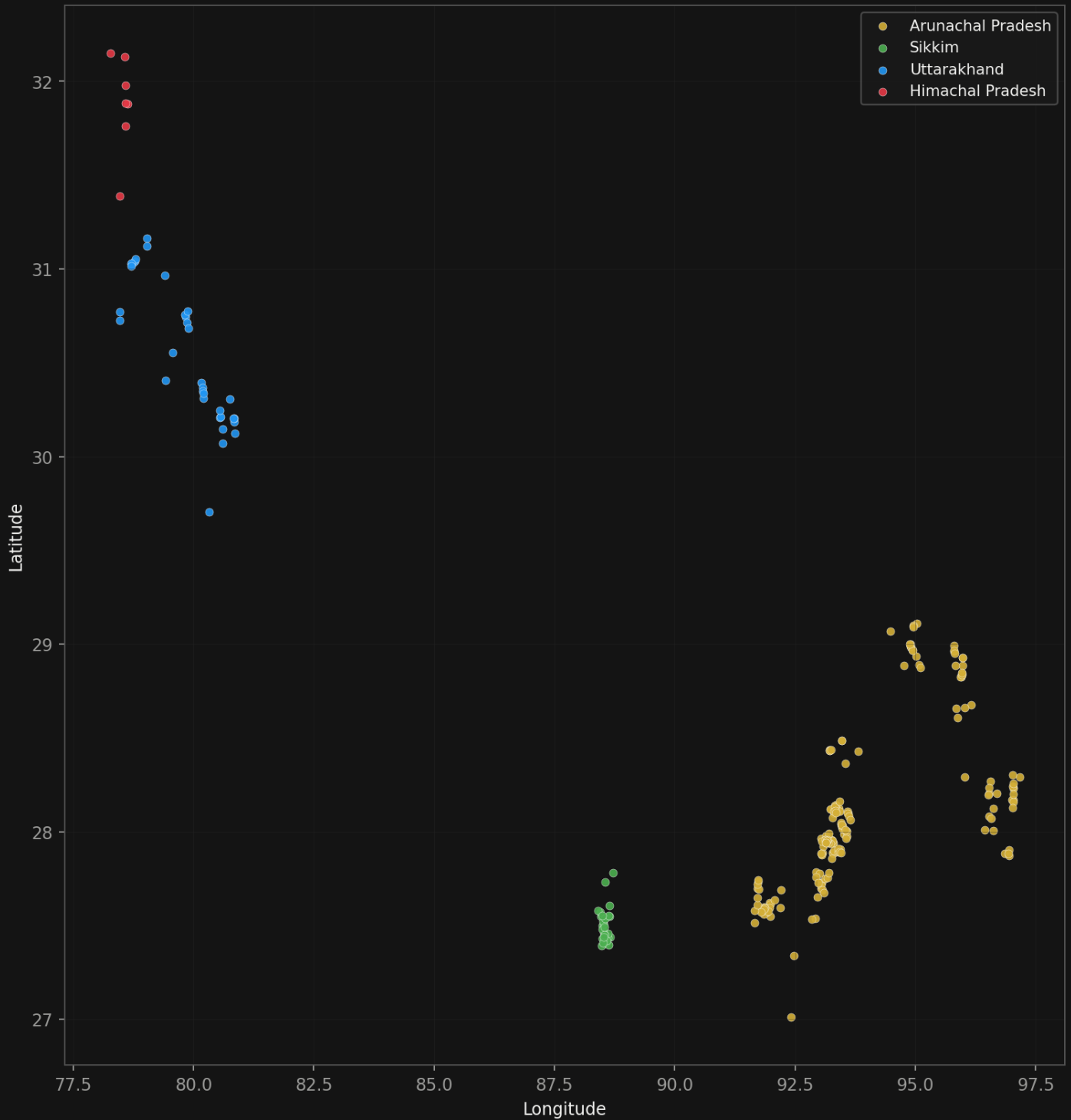
NDBI significance at 250m, 500m, and 1km buffer radii, summer window.

Figure 10 — Robustness Summary



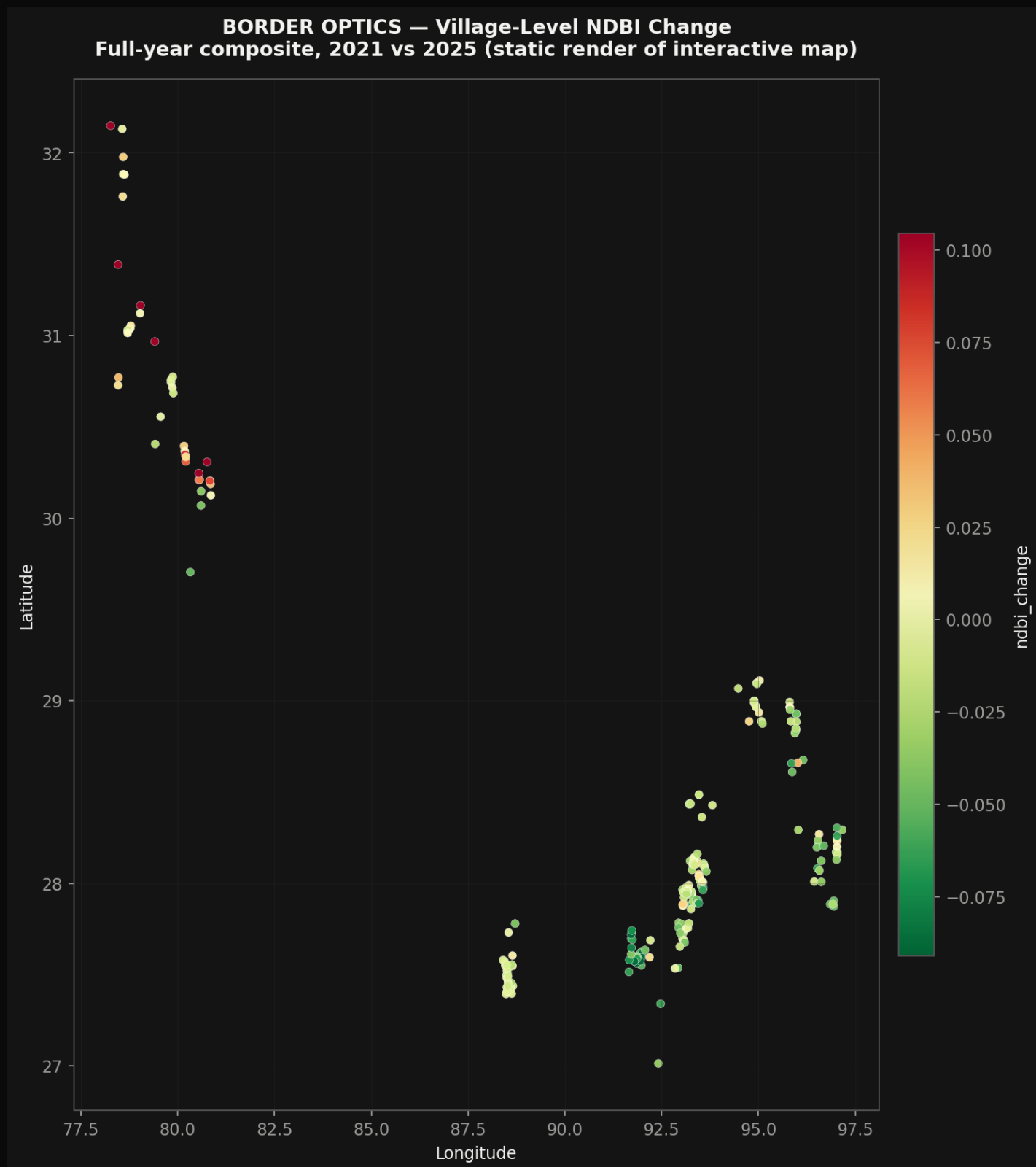
Map 1 — Village Sample Overview

BORDER OPTICS — Village Sample Overview
All geocoded villages across 4 states, colored by state (static render of interactive map)

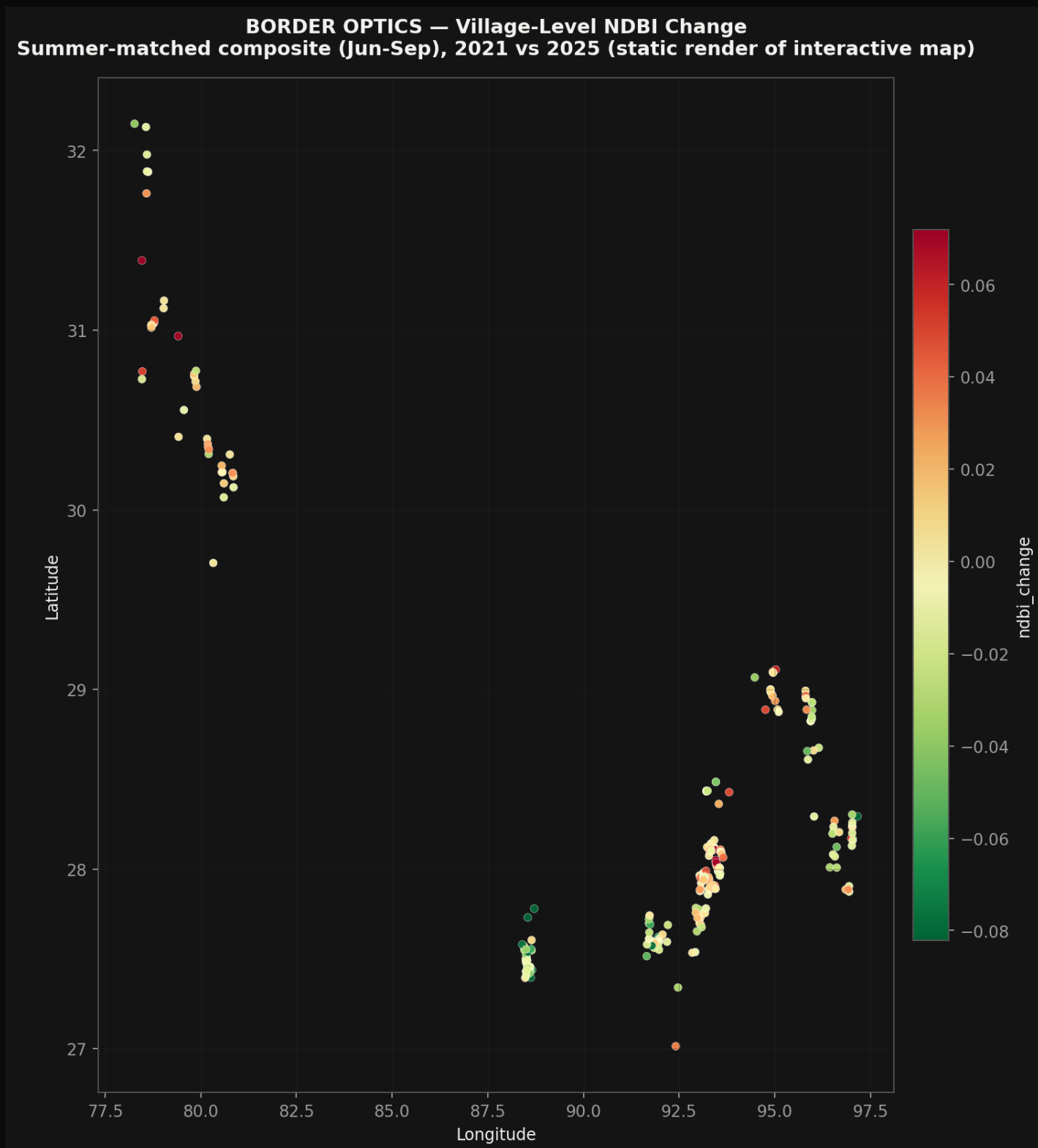


All 258 geocoded villages, colored by state. Interactive version: [dashboard](#) → [Interactive Maps](#) → [State Overview](#).

Map 2 — NDBI Change (Full-Year Composite)



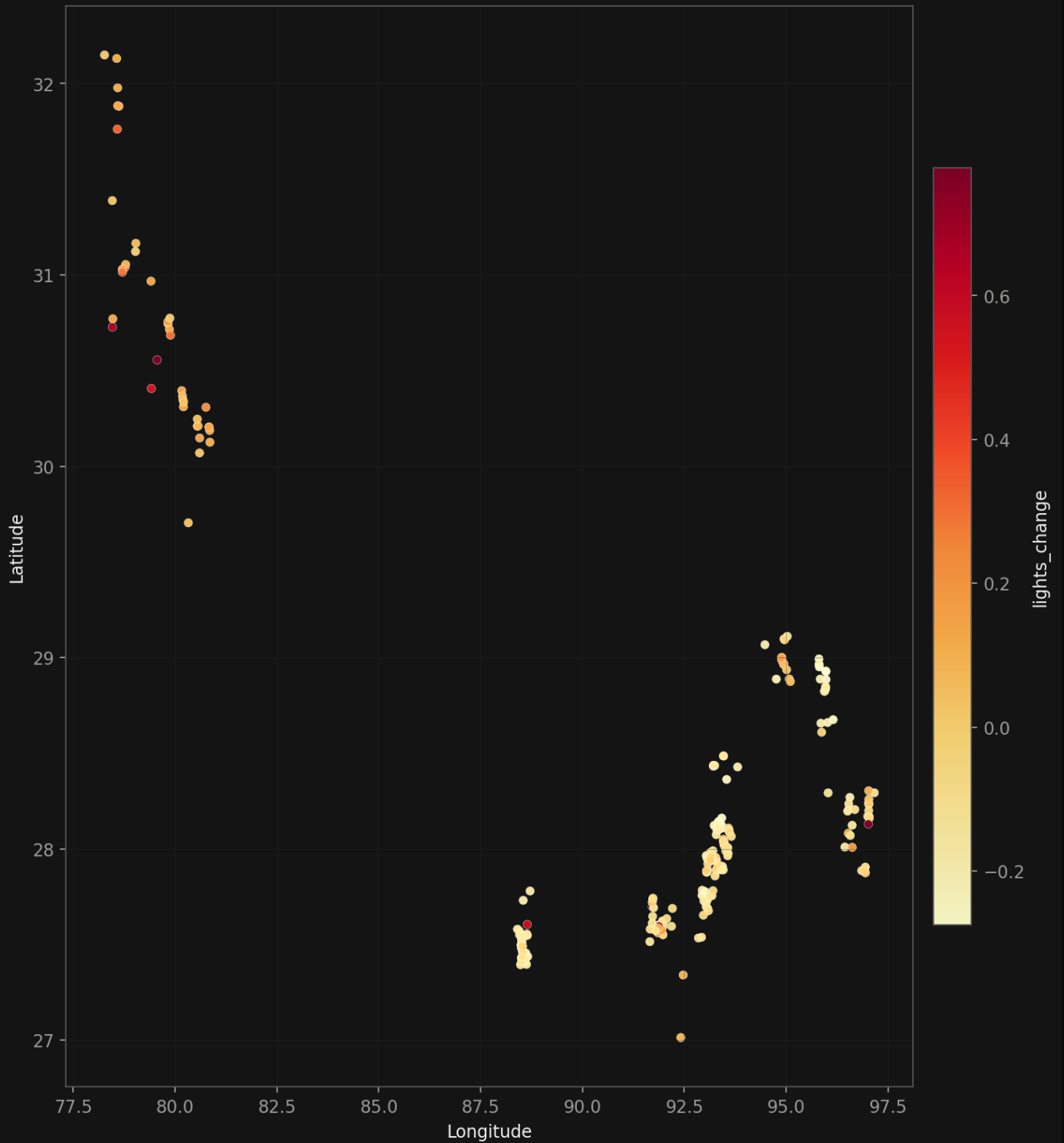
Map 3 — NDBI Change (Summer-Matched Composite)



Interactive version: [dashboard](#) → [Interactive Maps](#) → [Built-Up Change](#) — Summer-Matched Composite.

Map 4 — Night-Lights Change

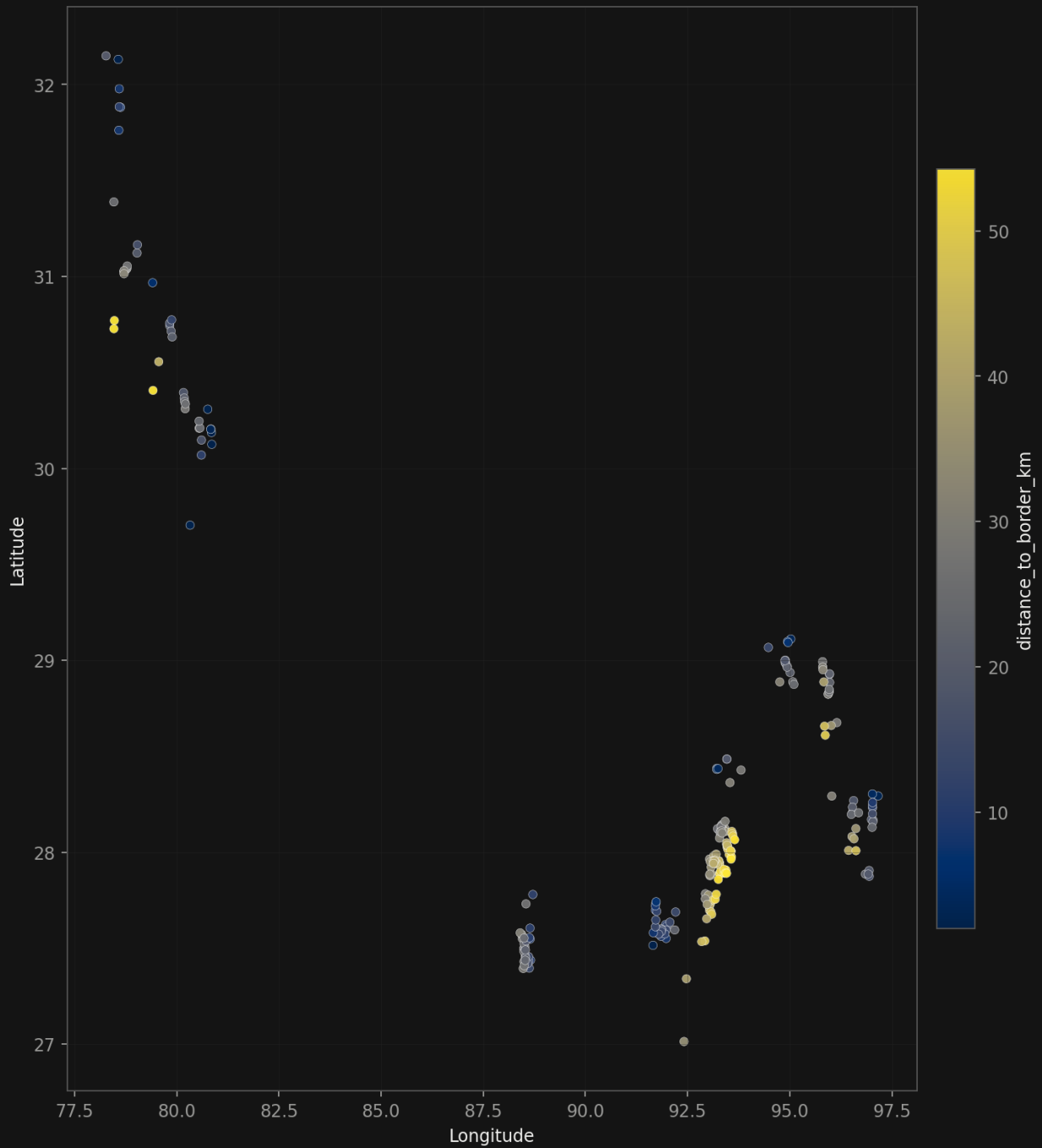
BORDER OPTICS — Village-Level Night-Lights Change
VIIRS radiance change, summer-matched composite (static render of interactive map)



Interactive version: [dashboard](#) → [Interactive Maps](#) → [Night-Lights Change](#).

Map 5 — Village Proximity to Border/LAC

BORDER OPTICS — Village Proximity to Border/LAC
Straight-line distance to nearest Natural Earth boundary segment (static render of interactive map)



Interactive version: [dashboard](#) → [Interactive Maps](#) → [Border / LAC Proximity](#).